Mathematics Revision Test 28

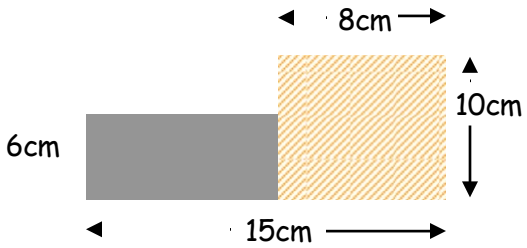
1. How much greater than $\frac{1}{2}$ is 0.64? _____

2. How much greater than $\frac{1}{2}$ is 0.502? _____

3. What is the time $\frac{3}{4}$ of an hour later than 23:20? _____

4. Find the difference between $\frac{3}{4}$ of £1 and 0.8 of £1 _____

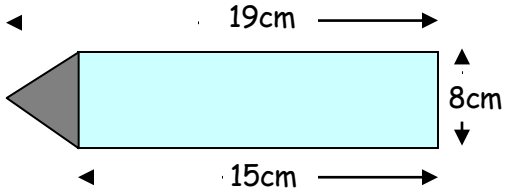
5. Find the **Total AREA** of this shape: _____



- Remember to find the area of the two shapes and add them together.

6. 85ml x 100 = _____ litres

7. Find the **Total AREA** of this shape: _____



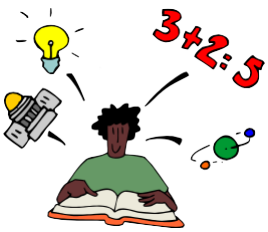
- Find the area of the rectangle first, then the area of the triangle and add them together. Do you remember how to find the area of a triangle?
- Answer: Half the base x the height.

8. $10^3 - 10^2$ = _____

9. 1% of sixteen thousand = _____

10. £1.75 x 5 = _____

- Please do any rough working out on this sheet.



Calculation Ladders

Same as last week, only a bit tougher!
You should be getting *quite fast* at these by now.
A reminder:
$\sqrt{n}$ means the square root of the number. Example: If n is 25, $\sqrt{25} = 5$

Ladder 6

1.	49
2.	$\sqrt{n}$
3.	X 6
4.	Find 50%
5.	Add 7
6.	Find 25%
7.	Multiply by 8
8.	Add on 4
9.	50% of this
10.	Find $33\frac{1}{3}$ %
	Your answer is

Ladder 7

1.	40
2.	75% of this
3.	Find $33\frac{1}{3}$ %
4.	Times 8
5.	50% of this
6.	25% of this
7.	n^2
8.	Divide by 4
9.	x 8
10.	Find 50%
	Your answer is

Are you enjoying these?

Equivalent Fractions

There is a very simple rule for Equivalent Fractions:

What you do to the top you must do to the bottom.
What you do to the bottom you must do the top.

Here the top numbers have been multiplied by 2

$$\overbrace{\frac{1}{4} = \frac{2}{8}}^{\times 2}$$

$$\underbrace{\frac{1}{4} = \frac{2}{8}}_{\times 2}$$

I must do the same to the bottom

$\frac{1}{2} = \frac{5}{10}$ Here both sets of numbers have been x by 5

$\frac{3}{4} = \frac{6}{8}$ Here both sets of numbers have been x by 2

Got it?

Now complete the exercise on the following sheet.

You will need to concentrate!

Equivalent fractions

Look at this row of fractions:

$\frac{1}{2}$ =	$\frac{5}{10}$	$\frac{6}{12}$	$\frac{2}{4}$	$\frac{4}{8}$	$\frac{3}{6}$

It contains 5 fractions, in black, each of which is equal in value or **EQUIVALENT** to the fraction in red.

The rule is: **What you do to the top you must do to the bottom.** SIMPS!

Set your work out like this in your exercise books:

$$\frac{1}{2} \xrightarrow{\times 5} = \frac{5}{10} \qquad \frac{1}{2} \xrightarrow{\times 6} = \frac{6}{12}$$

x5 x6

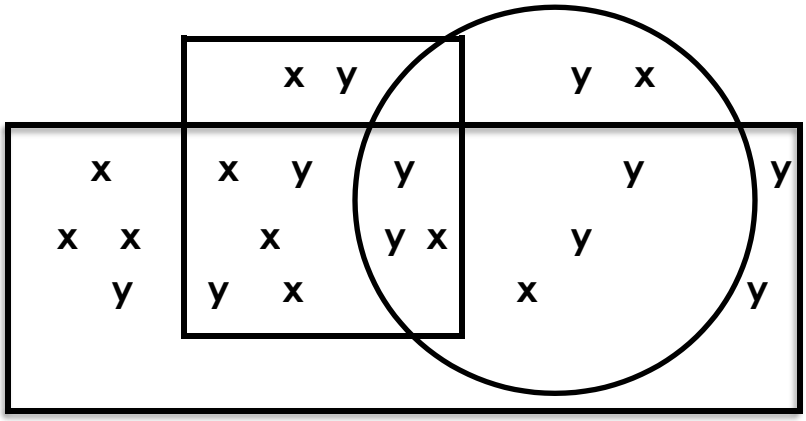
and so on.

Now complete the five fractions in each row below so they are all equivalent to the fraction in red.
Take your time. Complete this exercise nice and slowly. Enjoy it!

$\frac{1}{2}$ =	$\frac{_}{24}$	$\frac{_}{8}$	$\frac{3}{_}$	$\frac{_}{16}$	$\frac{_}{20}$
$\frac{1}{3}$ =	$\frac{6}{_}$	$\frac{2}{_}$	$\frac{3}{_}$	$\frac{_}{12}$	$\frac{_}{15}$
$\frac{1}{5}$ =	$\frac{6}{_}$	$\frac{_}{10}$	$\frac{_}{25}$	$\frac{_}{20}$	$\frac{3}{_}$
$\frac{2}{3}$ =	$\frac{12}{_}$	$\frac{_}{6}$	$\frac{10}{_}$	$\frac{_}{9}$	$\frac{8}{_}$
$\frac{3}{4}$ =	$\frac{18}{_}$	$\frac{6}{_}$	$\frac{9}{_}$	$\frac{_}{16}$	$\frac{_}{20}$
$\frac{1}{6}$ =	$\frac{6}{_}$	$\frac{2}{_}$	$\frac{5}{_}$	$\frac{_}{18}$	$\frac{_}{24}$
$\frac{2}{5}$ =	$\frac{_}{30}$	$\frac{4}{_}$	$\frac{_}{25}$	$\frac{6}{_}$	$\frac{_}{20}$
$\frac{5}{6}$ =	$\frac{_}{36}$	$\frac{_}{12}$	$\frac{25}{_}$	$\frac{15}{_}$	$\frac{20}{_}$
$\frac{4}{5}$ =	$\frac{12}{_}$	$\frac{_}{10}$	$\frac{_}{30}$	$\frac{20}{_}$	$\frac{16}{_}$
$\frac{3}{5}$ =	$\frac{18}{_}$	$\frac{6}{_}$	$\frac{_}{25}$	$\frac{12}{_}$	$\frac{15}{_}$

$\frac{1}{8}$ =	$\frac{2}{_}$	$\frac{_}{48}$	$\frac{5}{_}$	$\frac{4}{_}$	$\frac{_}{24}$
$\frac{1}{7}$ =	$\frac{_}{42}$	$\frac{_}{14}$	$\frac{5}{_}$	$\frac{3}{_}$	$\frac{_}{28}$
$\frac{3}{8}$ =	$\frac{_}{48}$	$\frac{6}{_}$	$\frac{15}{_}$	$\frac{_}{24}$	$\frac{_}{32}$
$\frac{1}{9}$ =	$\frac{6}{_}$	$\frac{2}{_}$	$\frac{3}{_}$	$\frac{_}{45}$	$\frac{_}{36}$
$\frac{5}{8}$ =	$\frac{30}{_}$	$\frac{10}{_}$	$\frac{_}{40}$	$\frac{15}{_}$	$\frac{_}{32}$
$\frac{4}{9}$ =	$\frac{24}{_}$	$\frac{_}{18}$	$\frac{20}{_}$	$\frac{_}{27}$	$\frac{16}{_}$
$\frac{7}{8}$ =	$\frac{_}{48}$	$\frac{_}{16}$	$\frac{35}{_}$	$\frac{_}{24}$	$\frac{28}{_}$
$\frac{3}{10}$ =	$\frac{18}{_}$	$\frac{_}{20}$	$\frac{_}{50}$	$\frac{9}{_}$	$\frac{12}{_}$
$\frac{8}{9}$ =	$\frac{48}{_}$	$\frac{16}{_}$	$\frac{_}{45}$	$\frac{_}{27}$	$\frac{_}{36}$
$\frac{7}{10}$ =	$\frac{42}{_}$	$\frac{_}{20}$	$\frac{_}{50}$	$\frac{_}{30}$	$\frac{28}{_}$

1. Study the diagram and answer the questions below:



a. How many x's and y's are there in total? _____

b. How many y's are only in the circle? _____

c. How many y's are in all three shapes? _____

d. How many x's are in the rectangle altogether? _____

e. How many y's are in the square and/or the rectangle only?_____

---

2. Multiply the number which is shown by 3 and add 7 to the answer. The first one has been done for you.

a.	***12***	**=**	***43***	**g.**	33	=	_____
b.	9	=	_____	**h.**	100	=	_____
c.	7	=	_____	**i.**	50	=	_____
d.	12	=	_____	**j.**	22	=	_____
e.	14	=	_____	**k.**	27	=	_____
f.	19	=	_____	**l.**	37	=	_____

3. Answer the following worded maths problems.

a. Add £33.87 to £43.95 then subtract £9.42. _____

b. What is the sum of £16.84 and £4? _____

c. What is left from £16.84 having spent £4.26? _____

d. I had £50 which I spent equally on presents for 4 cousins, how much did I spend on each? _____

e. If a train ticket costs £250 a month, how much would it cost to travel for a year? _____

f. I went to the optician. He charged £25.56 for an examination. How much change from £50 did I get? _____

g. A packet of bird seed costs 98p. How much does it cost to feed some canaries for a year if they eat one packet a month? _____

h. If I get a pay rise of 5% and my monthly salary was £3000, how much extra would I get a year? _____

i. If the exchange rate is 1.2 Euros to the £1, how much holiday money would I have in Euros if I had £50? _____

j. How much change would I get from £20 if I bought a bus ticket for £4.85? _____

4. Insert one of the missing signs in each space to make both sides equal.

× − + ÷

a. 6 ___ 6 = 6 ___ 2

b. 2 ___ 4 = 7 ___ 1

c. 9 ___ 4 = 60 ___ 24

d. 2 ___ 4 ___ 2 = 10

e. 3 ___ 6 ___ 1 = 17

f. 8 ___ 2 ___ 4 = 16

5. This box has numbers missing. Using the numbers 1-9 once only fill in the spaces so that each row, column and diagonal adds up to 15.

a.	**7**	b.
1	c.	d.
8	e.	f.

a. _____

b. _____

c. _____

d. _____

e. _____

f. _____

---

6. This is a bus timetable between the towns of Totton and Southampton. Use the information to answer the questions. Write your answers in the spaces below.

	A	B	C	D	E	F	G
Totton	**11.00**	**12.45**	**1.00**	**2.40**	**2.55**	**4.20**	**4.35**
Southampton	**11.30**	**11.45**	**1.40**	**1.55**	**3.30**	**3.45**	**5.50**

a. What bus takes the shortest time? _____

b. Which is the slowest? _____

c. What is the average time of journeys A, B, C, D and E? _____

d. If the schedule was continued when would the next bus leave Southampton? _____

e. If I miss the 1 o'clock to Southampton how long must I wait for the next bus? _____

f. Which two journeys take the same time? _____ and _____

7. Calculate the answers to the following:

a. 19 – 5 2/3 = _____

b. 18 – 20 + 57 = _____

c. 0.206 x 100 = _____

d. 6.73 x 100 = _____

e. 48/6 + 5 = _____

f. 15 1/4 + 8 $\frac{1}{2}$ = _____

g. 3 1/3 + 2 $\frac{1}{2}$ = _____

h. 742/7 = _____

i. 4+9+8–2+5–12 = _____

j. 16 x 7 = _____

k. 2.04 x 1000 = _____

l. 32.346 x 1000 = _____

m. 0.04 x 10 = _____

---

8. Place a ✓ in the space if the statement is true and a x if it is false.

a. There are one hundred minutes in one and a half hours. _____

b. There are 1000cm in a metre _____

c. A heptagon has eight sides. _____

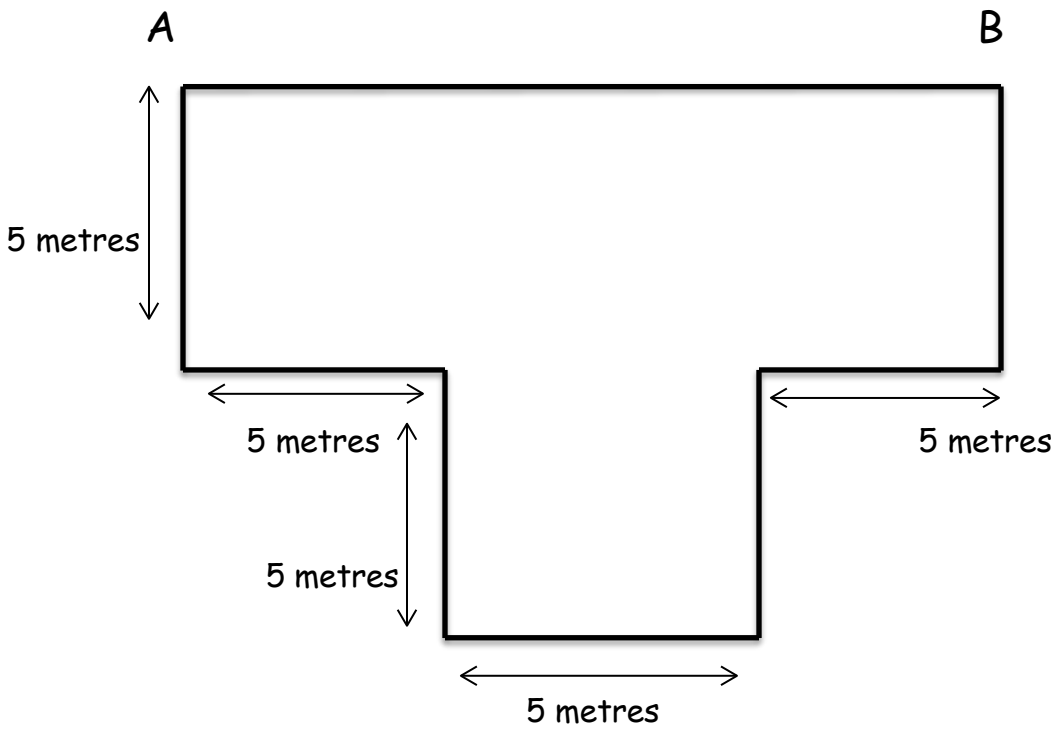
d. Seventy four divided by eight is nine. _____

e. There are 1000 millilitres in one litre. _____

f. The square root of 49 is 7. _____

g. Half of one hundred and thirty nine is sixty seven. _____

9. Study this diagram and then answer the questions below.



a. What is the length A-B in millimetres? _____

b. What is the perimeter of the shape in metres? _____

c. What is the area of the shape? _____

d. If the area is covered with a wooden floor at £6.50 per square metre how much would it cost? _____

e. If a border two metres wide was added all around the shape what would the new perimeter be? _____

f. If I have square tiles whose sides measure 50cm how many will I need to tile the whole space without any border? _____

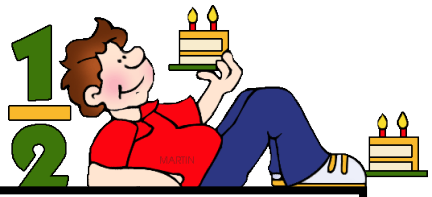
END OF TEST

If you have time

- Go back and complete any questions you missed out.

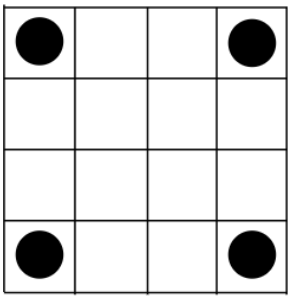
- Check your answers carefully.

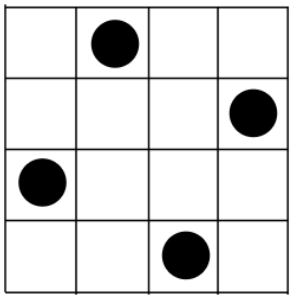
Maths Reasoning

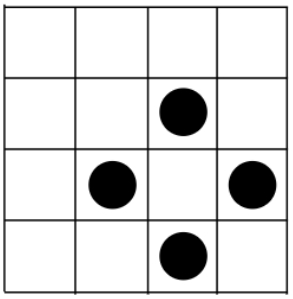


These questions are designed to get your mind whirring - so have a go. If you can't do it first time then try, try again. It is important to persevere.

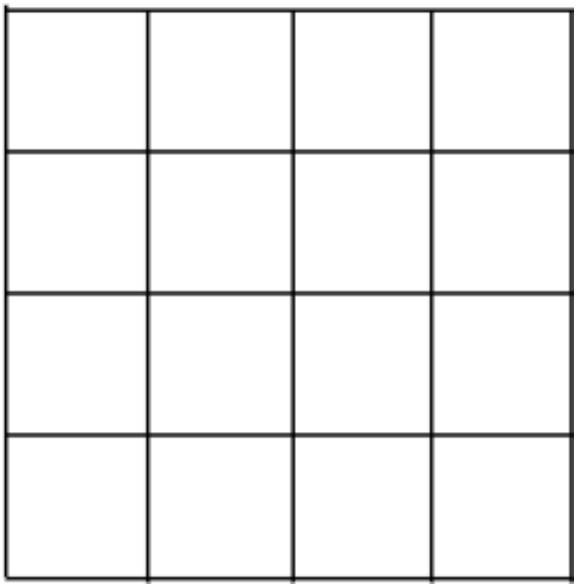
On each of the grids below there are 4 counters making up the corners of a square.







How many counters can you place on this grid without any of the counters making the four corners of a square?



Is there more than one way to arrange the maximum amount of counters?

ANSWERS - COMPLETE MATHS – PART 30

Maths

Revision Test

1. 0.14
2. 0.002
3. 00:05
1. 5p
2. 122 cm^2
3. 8.5 litres
4. 136 cm^2
5. 900
6. 160
7. £8.75

Calculation Ladders

Ladder 1 = 10

Ladder 2 = 100

Equivalent fractions

(Answers from top to bottom, left to right)

$\frac{1}{2}$ =	$\frac{12}{24}$	$\frac{4}{8}$	$\frac{3}{6}$	$\frac{8}{16}$	$\frac{10}{20}$
$\frac{1}{3}$ =	$\frac{6}{18}$	$\frac{2}{6}$	$\frac{3}{9}$	$\frac{4}{12}$	$\frac{5}{15}$
$\frac{1}{5}$ =	$\frac{6}{30}$	$\frac{2}{10}$	$\frac{5}{25}$	$\frac{4}{20}$	$\frac{3}{15}$
$\frac{2}{3}$ =	$\frac{12}{18}$	$\frac{4}{6}$	$\frac{10}{15}$	$\frac{6}{9}$	$\frac{8}{12}$
$\frac{3}{4}$ =	$\frac{18}{24}$	$\frac{6}{8}$	$\frac{9}{12}$	$\frac{12}{16}$	$\frac{15}{20}$
$\frac{1}{6}$ =	$\frac{6}{36}$	$\frac{2}{12}$	$\frac{5}{30}$	$\frac{3}{18}$	$\frac{4}{24}$
$\frac{2}{5}$ =	$\frac{12}{30}$	$\frac{4}{10}$	$\frac{10}{25}$	$\frac{6}{15}$	$\frac{8}{20}$
$\frac{5}{6}$ =	$\frac{30}{36}$	$\frac{10}{12}$	$\frac{25}{30}$	$\frac{15}{18}$	$\frac{20}{24}$
$\frac{4}{5}$ =	$\frac{12}{15}$	$\frac{8}{10}$	$\frac{24}{30}$	$\frac{20}{25}$	$\frac{16}{20}$
$\frac{3}{5}$ =	$\frac{18}{30}$	$\frac{6}{10}$	$\frac{15}{25}$	$\frac{12}{20}$	$\frac{15}{25}$
$\frac{1}{8}$ =	$\frac{2}{16}$	$\frac{6}{48}$	$\frac{5}{40}$	$\frac{4}{32}$	$\frac{3}{24}$
$\frac{1}{7}$ =	$\frac{6}{42}$	$\frac{2}{14}$	$\frac{5}{35}$	$\frac{3}{21}$	$\frac{4}{28}$
$\frac{3}{8}$ =	$\frac{18}{48}$	$\frac{6}{16}$	$\frac{15}{40}$	$\frac{9}{24}$	$\frac{12}{32}$
$\frac{1}{9}$ =	$\frac{6}{54}$	$\frac{2}{18}$	$\frac{3}{27}$	$\frac{5}{45}$	$\frac{4}{36}$
$\frac{5}{8}$ =	$\frac{30}{48}$	$\frac{10}{16}$	$\frac{25}{40}$	$\frac{15}{24}$	$\frac{20}{32}$
$\frac{4}{9}$ =	$\frac{24}{54}$	$\frac{8}{18}$	$\frac{20}{45}$	$\frac{12}{27}$	$\frac{16}{36}$
$\frac{7}{8}$ =	$\frac{42}{48}$	$\frac{14}{16}$	$\frac{35}{40}$	$\frac{21}{24}$	$\frac{28}{32}$
$\frac{3}{10}$ =	$\frac{18}{60}$	$\frac{6}{20}$	$\frac{15}{50}$	$\frac{9}{30}$	$\frac{12}{40}$
$\frac{8}{9}$ =	$\frac{48}{54}$	$\frac{16}{18}$	$\frac{40}{45}$	$\frac{24}{27}$	$\frac{32}{36}$
$\frac{7}{10}$ =	$\frac{42}{60}$	$\frac{14}{20}$	$\frac{35}{50}$	$\frac{21}{30}$	$\frac{28}{40}$

ANSWERS - COMPLETE MATHS – PART 30

Maths
type 1
paper 5

Page 1

1.
a. 21
b. 1
c. 2
d. 8
e. 6

2.
a. 43
b. 34
c. 28
d. 43
e. 49
f. 64
g. 106
h. 307
i. 157
j. 73
k. 88
l. 118

Page 2

1.
a. £68.40
b. £20.84
c. £12.58
d. £12.50
e. £3,000
f. £24.44
g. £11.76
h. £1,800
i. €60.00
j. £15.15

2.
a. + x
b. + - or x +
c. x -
d. x +
e. x -
f. ÷ x

Page 3

3.
a. 6
b. 2
c. 5
d. 9
e. 3
f. 4

6.
a. A
b. G
c. 42mins
d. 6.05
e.1hr40mins
f. E & F

Page 4

7.
a. 13 1/3
b. 55
c. 20.6
d. 673
e. 13
f. 23 3/4
g. 5 5/6
h. 106
i. 12
j. 112
k. 2040
l. 32346
m. 0.4

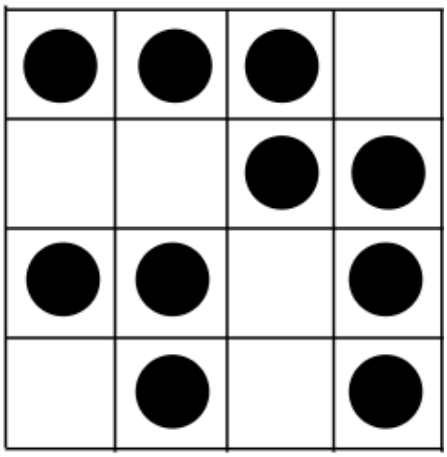
8.
a. x
b. x
c. x
d. x
e. ✓
f. ✓
g. x

Page 5

9.
a. 15000mm
b. 50m
c. 100m²
d. £650
e. 66m
f. 400

Maths Reasoning 20

The maximum number of counters you can place on the grid is 10.

There are multiple ways of doing this.

Here's one example: